■ Developer Mock Test Results

Test ID:	174cd931-4b40-4d...	**Student ID:**	7632
Test Type:	Developer Assessment	**Date:**	2026-01-20 11:17:37
Overall Score:	1/25 (4.0%)	**Performance:**	■■ Needs Improvement
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	1/10	10.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (1/10 - 10.0%)

■ Q1. Two people, A and B, can do a piece of work in 10 days and 15 days respectively. How many days will it take for A and B to do the work together?

Your Answer: 6 days

■ *Excellent! Your mathematical approach was correct.*

■ Q2. A mixture is made of two liquids A and B in the ratio 2:3. If the quantity of liquid A is 24 liters, what is the quantity of liquid B?

Your Answer: __SKIPPED__

Correct Answer: 36 liters

■ *The correct answer is 36 liters because the ratio of liquid A to liquid B is 2:3, and if the quantity of liquid A is 24 liters, we can find the quantity of liquid B by multiplying 24 by 3/2, which equals 36. If you skipped this question or got it wrong, you might have misunderstood how to apply the ratio to find the quantity of liquid B, but don't worry, it's an easy concept to grasp with a little practice. Remember to always use the given ratio to set up a proportion and solve for the unknown quantity, and you'll become more confident in your problem-solving skills.*

■ Q3. The average of 10 numbers is 15. If one of the numbers is 20, what is the sum of the remaining 9 numbers?

Your Answer: __SKIPPED__

Correct Answer: 130

■ *The correct answer is 130 because the average of 10 numbers is 15, meaning the total sum of the numbers is 10 * 15 = 150. If one of the numbers is 20, then the sum of the remaining 9 numbers is 150 - 20 = 130. If you skipped this question or got it wrong, you might have forgotten to subtract the given number from the total sum, so remember to always read the question carefully and apply the correct operations to find the solution.*

■ **Q4. There are five houses, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different nationa...**

Your Answer: White
Correct Answer: Blue
■ *The correct answer is Blue because the Canadian lives in the first house and since the American lives in the red house, and there's no indication that the Canadian's house is red, we can deduce the Canadian's house must be a different colour, and given the other clues, Blue is the only possibility. You likely made a mistake by not fully considering the implications of the Canadian living in the first house and the American living in the red house, which led you to incorrectly choose White. Don't worry, it's all part of the learning process, and with practice, you'll get better at solving these types of puzzles.*

■ **Q5. What is the next number: 3, 6, 9, 12, 15, ?**

Your Answer: __SKIPPED__
Correct Answer: 18
■ *The correct answer is 18 because the pattern in the sequence is adding 3 to the previous number, so 3 + 3 = 6, 6 + 3 = 9, and so on. If you skipped this question, you might have missed the simple addition pattern, but that's okay - it's all part of the learning process. Keep practicing and you'll become more confident in recognizing these types of patterns, and you'll be able to tackle similar questions with ease.*

■ **Q6. What is the next number: 2, 6, 12, 20, 30, ?**

Your Answer: __SKIPPED__
Correct Answer: 42
■ *The correct answer is 42 because the pattern in the sequence is formed by adding 4, 6, 8, 10, and so on, increasing the increment by 2 each time. This sequence is a classic example of a quadratic or second-order sequence, where each term increases by consecutive even numbers. If you chose a different answer, you might have misidentified the pattern or misunderstood how the increments were changing, but don't worry, recognizing and learning from these patterns takes practice, so keep exploring and you'll get better at identifying them.*

■ **Q7. If price increases from Rs.200 to Rs.250, what is the percentage increase?**

Your Answer: __SKIPPED__
Correct Answer: 25%
■ *The correct answer is 25% because the price increase from Rs.200 to Rs.250 is Rs.50, and to find the percentage increase, we calculate (increase/original price) * 100, which is (50/200) * 100 = 25%. If you chose a different option, you might have made a calculation error, such as incorrectly finding the increase or dividing by the wrong number. Don't worry, it's an easy mistake to make, and with practice, you'll become more confident in calculating percentage increases.*

■ **Q8. A shopkeeper buys 100 items at Rs.10 each and sells 50 items at Rs.15 each. What is the percentage profit?**

Your Answer: __SKIPPED__
Correct Answer: 25%
■ *To find the percentage profit, first, calculate the cost price and selling price of the items sold. The shopkeeper buys 100 items at Rs.10 each, so the cost price for 50 items is Rs.500, and selling 50 items at Rs.15 each gives a selling price of Rs.750, resulting in a profit of Rs.250, which is 25% of the cost price, hence the correct answer is 25%. If the user was wrong, they might have calculated the cost price or selling price incorrectly, or misunderstood the concept of percentage profit, but with practice and review, they can master this concept and improve their calculations.*

■ **Q9. A pipe can fill a tank in 6 hours. Another pipe can empty the tank in 8 hours. How long will it take to fill the tank if both pipes are open?**

Your Answer: __SKIPPED__
Correct Answer: 24 hours
■ *The correct answer is 24 hours because when both pipes are open, the net rate of filling the tank is the difference between the rates at which each pipe fills or empties the tank. If the user skipped this question or got it wrong, they might have misunderstood how to calculate the combined rate of the two pipes, which is 1/6 (fill rate) - 1/8 (empty rate) = 1/24, meaning the tank fills at a rate of 1/24 per hour, taking 24 hours to fill. This is a great opportunity to review and practice working with rates and ratios to build problem-solving skills.*

■ **Q10. A mixture is made of two liquids A and B in the ratio 3:5. If the quantity of liquid A is 36 liters, what is the quantity of liquid B?**

Your Answer: __SKIPPED__
Correct Answer: 60 liters

■ *The correct answer is 60 liters because the ratio of liquid A to liquid B is 3:5, and if the quantity of liquid A is 36 liters, we can find the quantity of liquid B by setting up a proportion and solving for it, which results in 60 liters. If you skipped this question or got it wrong, you might have misunderstood how to apply the ratio to find the unknown quantity, so take a closer look at how ratios work and practice applying them to similar problems. Remember, ratios are all about proportions, so make sure to set up your proportions correctly and solve for the unknown value.*

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is the primary benefit of using object-oriented programming in Python?

Your Answer: __SKIPPED__
Correct Answer: Reduced code duplication and improved program organization
■ *The correct answer is right because object-oriented programming in Python allows for the creation of reusable code blocks, such as classes and objects, which reduces code duplication and improves program organization. This makes it easier to maintain and modify code, making it a fundamental benefit of object-oriented programming. If you skipped this question, don't worry, it's a great opportunity to learn and understand the basics of object-oriented programming, and with practice, you'll become more familiar with its benefits and applications.*

■ Q2. What are the four fundamental principles of object-oriented programming in Python?

Your Answer: __SKIPPED__
Correct Answer: Encapsulation, abstraction, inheritance, and polymorphism
■ *The correct answer is encapsulation, abstraction, inheritance, and polymorphism because these four principles form the foundation of object-oriented programming in Python, allowing for effective organization, reuse, and modification of code. If you skipped this question or chose a different option, you might have been unsure about the distinction between related concepts like composition, overloading, and overriding, which are important but not part of the core four principles. Don't worry, it's all part of the learning process, and understanding these principles will help you become proficient in Python and object-oriented programming.*

■ Q3. What is the correct step to create a class in Python?

Your Answer: __SKIPPED__
Correct Answer: Define a class using the class keyword
■ *To create a class in Python, you need to use the class keyword, which is a specific keyword reserved for defining classes. If you skipped this question or chose a different option, you might have gotten confused with other fundamental concepts in Python, such as defining functions or importing modules, but the class keyword is unique to creating classes. Don't worry, it's an easy concept to grasp, and with practice, you'll become more comfortable with the basics of Python programming.*

■ Q4. What is the purpose of the __init__ method in Python classes?

Your Answer: __SKIPPED__
Correct Answer: To initialize objects with instance variables
■ *The correct answer is right because the __init__ method in Python classes is a special method that is automatically called when an object of the class is created, allowing you to initialize objects with instance variables. If you skipped this question, you might have been unsure about the purpose of the __init__ method, but don't worry, it's a key concept in object-oriented programming that you can easily learn and practice. By understanding the __init__ method, you'll be able to create classes that can effectively store and manage data, so keep exploring and practicing Python to become more confident in your coding skills.*

■ Q5. What is the purpose of method overriding in Python classes?

Your Answer: __SKIPPED__
Correct Answer: To allow child classes to provide their own implementation of a method
■ *Method overriding in Python classes is used to allow child classes to provide their own implementation of a method, which is why option C is the correct answer. This is a fundamental concept in object-oriented programming, enabling more specific behavior in subclasses while maintaining a common interface. If you skipped this question, don't worry - it's a great opportunity to learn and understand this key concept, and with practice, you'll become more confident in your ability to apply method overriding in your Python programming.*

■ Q6. What is the primary characteristic of the Python programming language?

Your Answer: __SKIPPED__
Correct Answer: It is a high-level, interpreted language

■ *The correct answer is that Python is a high-level, interpreted language because it provides a simple syntax and doesn't require manual memory management, making it easy to learn and use, and its code is executed line by line at runtime. If you skipped this question or chose a different option, you might have misunderstood the key features of Python, but that's okay - it's all part of the learning process. Remember, Python's interpretive nature and high-level abstractions are what make it a popular and versatile language for beginners and experienced programmers alike.*

■ Q7. What are the fundamental concepts of Python programming?

Your Answer: __SKIPPED__

Correct Answer: Variables, data types, control structures, and functions

■ *The correct answer is right because variables, data types, control structures, and functions are the building blocks of Python programming, providing the foundation for writing efficient and effective code. If the user skipped this question, they might have overlooked the importance of these fundamental concepts, which are essential for any beginner in Python programming. By understanding these basics, users can develop a strong foundation and confidently move on to more advanced topics, so it's great that they can review and learn from this question.*

■ Q8. What is the correct step to write a Python program?

Your Answer: __SKIPPED__

Correct Answer: Install Python, open a code editor, write code, run the program

■ *The correct answer is the right sequence because you need to install Python first to have the necessary software to run your program, then open a code editor to write your code, and finally run the program to see the results. If you skipped this question or chose a different option, you might have gotten the order mixed up, which is an easy mistake to make, but don't worry, it's all part of the learning process. By following the correct steps, you'll be able to write and run your Python programs successfully, so keep practicing and you'll get the hang of it.*

■ Q9. What is the purpose of functions in Python programming?

Your Answer: __SKIPPED__

Correct Answer: To organize code into reusable units

■ *The correct answer is right because functions in Python allow you to break down your code into smaller, manageable chunks that can be reused throughout your program, making it more efficient and easier to maintain. If you skipped this question, you might have overlooked the fundamental concept of functions as building blocks of code organization, which is essential for writing clean and scalable code. Don't worry, it's an easy concept to grasp, and with practice, you'll become proficient in using functions to streamline your Python programming.*

■ Q10. What is the purpose of using try, except, and finally blocks in Python programming?

Your Answer: __SKIPPED__

Correct Answer: To handle errors and exceptions during program execution

■ *The correct answer is right because try, except, and finally blocks in Python are used to handle errors and exceptions that may occur during program execution, allowing developers to gracefully manage and recover from unexpected events. If the user skipped this question, they may not have understood the importance of error handling in programming, which is a crucial aspect of writing robust and reliable code. By using these blocks, developers can ensure their programs remain stable and provide meaningful error messages, making it an essential skill to master in Python programming.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Python function to calculate the average grade of a student given their scores in three subjects: math, science, and English. **Input:** A dic...

Your Answer: def calculate_total_salary(base_salary, bonus_percentage): bonus_amount = (bonus_percentage / 100) * base_salary total_salary = base_salary + bonus_amount return total_salary base_salary = 5000 bonus_percentage = 10 total_salary = calculate_total_salary(base_salary, bonus_percentage) print("Base Salary: ", base_salary) print("Bonus Percentage: ", bonus_percentage, "%") print("Total Salary: ", total_salary)

Correct Answer: def calculate_average_grade(grades): """ Calculate the average grade of a student given their scores in three subjects. Args: grades (dict): A dictionary with subject names as keys and scores as values. Returns: float: The average grade of the student. """ # Check if the input dictionary has the required subjects required_subjects = ["math", "science", "english"] if set(required_subjects).issubset(grades.keys()): # Calculate the sum of the grades total_grade = sum(grades.values()) # Calculate the average grade average_grade = total_grade / len(grades) return round(average_grade, 2) else: return "Invalid input: Please provide grades for math, science, and english." # Example usage: grades = {"math": 85, "science": 90, "english": 78} average = calculate_average_grade(grades) print("Average grade:", average)

■ *The issue with the student's code is that it's attempting to calculate a total salary with a base salary and bonus percentage, which is unrelated to the task of calculating an average grade from a dictionary of subject scores. In contrast, the correct code works because it defines a function `calculate_average_grade` that takes a dictionary of grades as input, checks for the required subjects, and then calculates the average grade by summing the values and dividing by the number of subjects. By following this approach, the correct code provides a clear and effective solution to the problem, and with practice, you'll become proficient in writing similar functions to solve real-world problems.*

■ Q2. Write a Python program to count the number of lines in a given text file. **Input:** The name of the text file (e.g., "example.txt") **Output:** The n...

Your Answer: def calculate_total_salary(base_salary, bonus_percentage): bonus_amount = (bonus_percentage / 100) * base_salary total_salary = base_salary + bonus_amount return total_salary base_salary = 5000 bonus_percentage = 10 total_salary = calculate_total_salary(base_salary, bonus_percentage) print("Base Salary: ", base_salary) print("Bonus Percentage: ", bonus_percentage, "%") print("Total Salary: ", total_salary)

Correct Answer: def count_lines_in_file(filename): try: with open(filename, 'r') as file: lines = file.readlines() return len(lines) except FileNotFoundError: print(f"File '{filename}' not found.") return None filename = "example.txt" line_count = count_lines_in_file(filename) if line_count is not None: print(f"The file '{filename}' has {line_count} lines.")

■ *It looks like there's a bit of a mix-up in your code - you've written a function to calculate a total salary, but the task was to count the number of lines in a text file. The correct code works because it opens the specified file, reads all the lines, and then returns the count of those lines using the `len()` function. Don't worry, mistakes like this are a great learning opportunity, and with a little practice, you'll be writing file-reading code like a pro!*

■ Q3. Write a Python class to manage bank account transactions. The class should have methods to deposit, withdraw, and check the balance. **Input:** Initia...

Your Answer: def calculate_total_salary(base_salary, bonus_percentage): bonus_amount = (bonus_percentage / 100) * base_salary total_salary = base_salary + bonus_amount return total_salary base_salary = 5000 bonus_percentage = 10 total_salary = calculate_total_salary(base_salary, bonus_percentage) print("Base Salary: ", base_salary) print("Bonus Percentage: ", bonus_percentage, "%") print("Total Salary: ", total_salary)

Correct Answer: class BankAccount: def __init__(self, initial_balance): self.balance = initial_balance def deposit(self, amount): self.balance += amount def withdraw(self, amount): if amount > self.balance: print("Insufficient balance") else: self.balance -= amount def check_balance(self): return self.balance def main(): initial_balance = 100 transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)] account = BankAccount(initial_balance) for transaction in transactions: if transaction[0] == "deposit": account.deposit(transaction[1]) elif transaction[0] == "withdraw": account.withdraw(transaction[1]) print("Final balance:", account.check_balance()) if __name__ == "__main__": main()

■ *The issue with your code is that it seems to be solving a completely different problem related to salary calculations, rather than managing bank account transactions as requested. In contrast, the correct code works because it defines a `BankAccount` class with methods to deposit, withdraw, and check the balance, allowing it to accurately process transactions and calculate the final balance. By creating a class specifically designed for bank account management, the correct code is able to effectively solve the problem in a clear and organized manner.*

■ Q4. Write a Python program to download multiple files concurrently using threads. **Input:** A list of file URLs (e.g., `["http://example.com/file1.txt", ...

Your Answer: def calculate_total_salary(base_salary, bonus_percentage): bonus_amount = (bonus_percentage / 100) * base_salary total_salary = base_salary + bonus_amount return total_salary base_salary = 5000 bonus_percentage = 10 total_salary = calculate_total_salary(base_salary, bonus_percentage) print("Base Salary: ", base_salary) print("Bonus Percentage: ", bonus_percentage, "%") print("Total Salary: ", total_salary)

Correct Answer: import requests import threading def download_file(url): filename = url.split("/")[-1] response = requests.get(url) with open(filename, 'wb') as file: file.write(response.content) print(f"Downloaded {filename}") def main(): urls = ["http://example.com/file1.txt", "http://example.com/file2.txt", "http://example.com/file3.txt"] threads = [] for url in urls: thread = threading.Thread(target=download_file, args=(url,)) threads.append(thread) thread.start() for thread in threads: thread.join() print("All files downloaded") if __name__ == "__main__": main()

■ *The issue with your code is that it appears to be calculating a total salary based on a base salary and bonus percentage, which is unrelated to the task of downloading multiple files concurrently using threads. In contrast, the correct code works because it utilizes the `threading` module to create separate threads for each file download, allowing them to run concurrently, and the `requests` library to handle the actual file downloads. By using threads and a clear, focused approach, the correct code is able to efficiently download multiple files and provide a success message, so let's build on this example to explore more about concurrent programming in Python.*

■ Q5. Write a Python program to convert a CSV file to a JSON file. **Input:** The name of the CSV file (e.g., "data.csv") **Output:** The name of the result...

Your Answer: def calculate_total_salary(base_salary, bonus_percentage): bonus_amount = (bonus_percentage / 100) * base_salary total_salary = base_salary + bonus_amount return total_salary base_salary = 5000 bonus_percentage = 10 total_salary = calculate_total_salary(base_salary, bonus_percentage) print("Base Salary: ", base_salary) print("Bonus Percentage: ", bonus_percentage, "%") print("Total Salary: ", total_salary)

Correct Answer: import csv import json def csv_to_json(csv_file_name): # Read the CSV file with open(csv_file_name, 'r') as csv_file: csv_reader = csv.DictReader(csv_file) data = list(csv_reader) # Get the JSON file name by replacing '.csv' with '.json' json_file_name = csv_file_name.replace('.csv', '.json') # Write the data to the JSON file with open(json_file_name, 'w') as json_file: json.dump(data, json_file, indent=4) print(f"CSV file '{csv_file_name}' has been converted to JSON file '{json_file_name}'") # Example usage: csv_file_name = "data.csv" csv_to_json(csv_file_name) # Print the contents of the JSON file with open(csv_file_name.replace('.csv', '.json'), 'r') as json_file: print(json_file.read())

■ *The issue with your code is that it's attempting to calculate a total salary instead of converting a CSV file to a JSON file, which is the task at hand. The correct code works because it utilizes the `csv` and `json` modules in Python to read the CSV file and write its data to a JSON file, effectively converting the file format. Don't worry, mistakes like this are an opportunity to learn and improve - let's take a closer look at the correct code together to understand how it achieves the desired outcome!*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.